ADS1298 #1

Electrodes

E1 R1 2KΩ R2 2KΩ C1 620pF C2 620pF GND

E2 R3 2KΩ R4 2KΩ C3 620pF C4 620pF GND

E3 R5 2KΩ R6 2KΩ C5 620pF C10 620pF GND

E4 R7 2KΩ R8 2KΩ C11 620pF C12 620pF GND

E5 R9 2KΩ R10 2KΩ C13 620pF C14 620pF GND

E6 R11 2KΩ R12 2KΩ C15 620pF C16 620pF GND

E7 R13 2KΩ R14 2KΩ C17 620pF C18 620pF GND

E8 R15 2KΩ R16 2KΩ C21 620pF C22 620pF GND

+3.3V AVDD C6 0.1uF C7 10uF AGND

DVDD +3.3V C8 0.1uF C9 10uF AGND

U1 ADS1298xPAG

IN1P IN1N IN2P IN2N IN3P IN3N IN4P IN4N IN5P IN5N IN6P IN6N IN7P IN7N IN8P IN8N

AVDD AVDD1 DVDD

WCT RLDOUT RLDIN RLDINV RLDREF

AVDD

TESTP_PACE_OUT1 TESTP_PACE_OUT2

GPIO1 GPIO2 GPIO3 GPIO4 AGND

DAISY_IN SCLK DIN CS DOUT

Unused, shorted AGND

SCLK MOSI CS_ADC1 MISO

CLK CLKSEL START DRDY PWDN RESET RESV1

U1 CLK_EN=1 required

ADC_CLK DVDD START DRDY_ADC1 PWDN RESET AGND

VCAP1_ADC1 VCAP2_ADC1 VCAP3_ADC1 VCAP4_ADC1

VREFP_ADC1 C19 0.1uF C20 10uF

VREFP VREFN AVSS AVSS1 AGND DGND

C23 22uF C24 1uF C25 1uF C26 1uF AGND

ADS1298 #2

Electrodes

E9 R17 2KΩ R18 2KΩ C27 620pF C28 620pF GND

E10 R19 2KΩ R20 2KΩ C29 620pF C30 620pF GND

E11 R21 2KΩ R22 2KΩ C31 620pF C36 620pF GND

E12 R23 2KΩ R24 2KΩ C37 620pF C38 620pF GND

E13 R25 2KΩ R26 2KΩ C39 620pF C40 620pF GND

E14 R27 2KΩ R28 2KΩ C41 620pF C42 620pF GND

E15 R29 2KΩ R30 2KΩ C43 620pF C44 620pF GND

E16 R31 2KΩ R32 2KΩ C47 620pF C48 620pF GND

+3.3V AVDD C32 0.1uF C33 10uF AGND

DVDD +3.3V C34 0.1uF C35 10uF AGND

U2 ADS1298xPAG

IN1P IN1N IN2P IN2N IN3P IN3N IN4P IN4N IN5P IN5N IN6P IN6N IN7P IN7N IN8P IN8N

AVDD AVDD1 DVDD

WCT RLDOUT RLDIN RLDINV RLDREF

AVDD

TESTP_PACE_OUT1 TESTP_PACE_OUT2

GPIO1 GPIO2 GPIO3 GPIO4 AGND

DAISY_IN SCLK DIN CS DOUT

Unused, shorted AGND

SCLK MOSI CS_ADC2 MISO

CLK CLKSEL START DRDY PWDN RESET RESV1

U1 CLK_EN=1 required

ADC_CLK START DRDY_ADC2 PWDN RESET AGND

F09: DRDY_ADC2 only; MCU still uses U1 DRDY

VCAP1_ADC2 VCAP2_ADC2 VCAP3_ADC2 VCAP4_ADC2

VREFP_ADC2 C45 0.1uF C46 10uF

VREFP VREFN AVSS AVSS1 AGND DGND

C49 22uF C50 1uF C51 1uF C52 1uF AGND

+5V +3.3V DVDD

U3 ESP32-S3-DevKitC

CHIP_PU 5V 3V3

GPIO43/U0TXD GPIO44/U0RXD

GPIO15/ADC2_CH4/32K_P GPIO16/ADC2_CH5/32K_N

GPIO20/USB_D+ GPIO19/USB_D-

GPIO39/MTCK GPIO40/MTDO GPIO41/MTDI GPIO42/MTMS

GPIO0 GPIO1/ADC1_CH0 GPIO2/ADC1_CH1 GPIO3/ADC1_CH2 GPIO4/ADC1_CH3 GPIO5/ADC1_CH4 GPIO6/ADC1_CH5 GPIO7/ADC1_CH6 GPIO8/ADC1_CH7 GPIO9/ADC1_CH8 GPIO10/ADC1_CH9 GPIO11/ADC2_CH0 GPIO12/ADC2_CH1 GPIO13/ADC2_CH2 GPIO14/ADC2_CH3 GPIO17/ADC2_CH6 GPIO18/ADC2_CH7 GPIO21 GPIO35 GPIO36 GPIO37 GPIO38 GPIO45 GPIO46 GPIO47 GPIO48

ESP32-S3-DevKitC

RESET PWDN START DRDY_ADC1 CS_ADC1 MOSI SCLK MISO CS_ADC2

GND AGND

F21: probe access only; test-pad footprints/layout pending

TP1 ADC_CLK — ADC_CLK

TP2 DRDY_ADC1 — DRDY_ADC1

TP3 DRDY_ADC2 — DRDY_ADC2

TP4 3V3 — +3.3V

TP5 BAT — +3.7V

TP6 AGND — AGND

TP7 VREFP_ADC1 — VREFP_ADC1

TP8 VCAP1_ADC1 — VCAP1_ADC1

TP9 VCAP2_ADC1 — VCAP2_ADC1

TP10 VCAP3_ADC1 — VCAP3_ADC1

TP11 VCAP4_ADC1 — VCAP4_ADC1

TP12 VREFP_ADC2 — VREFP_ADC2

TP13 VCAP1_ADC2 — VCAP1_ADC2

TP14 VCAP2_ADC2 — VCAP2_ADC2

TP15 VCAP3_ADC2 — VCAP3_ADC2

TP16 VCAP4_ADC2 — VCAP4_ADC2

Grounding Configuration

AGND GND

ADC LDO Configuration

U4 LP2985-3.3

+3.7V C53 1uF GND

VIN ON/OFF EN GND BP VOUT

+3.3V C54 2.2uF GND

C55 10nF GND

Battery Configuration

+3.7V BT1 Battery GND

MCU LDO Configuration

U5 TLV75533PDBV

+3.7V C56 1uF GND

IN EN GND OUT

+3.3V C57 200uF GND

START R33 10k AGND

RESET R34 10k AGND

PWDN R35 10k AGND

F09: 10k pulldowns bias START, RESET and PWDN low while MCU is high-Z.
CS defaults / complete power sequencing remain unresolved.

MOSI R36 10k AGND

SCLK R37 10k

EN R38 0

V1.2 F09: R36/R37 bias DIN and SCLK low; CS / full sequencing unresolved.
F05: R38 = OR FITTED: both LDOs held OFF. No functional enable source.
Do not remove R38 or drive EN until F04/F13/F15/F16/F17 are resolved.
USB / external +3.3V can still backfeed the rail: OFF strap is not isolation.

V1.2 PARTIAL CORRECTIONS — NOT READY FOR POWER-UP OR WEARER USE
F04: LDO outputs remain parallel. F05: fitted R38 holds EN low; power architecture unresolved.
F06/F18: restore U1 CLK_EN after RESET; allow U2 clock/POR; configure before shared START.
F09: START/RESET/PWDN plus DIN/SCLK have pulldowns; CS/full sequencing unresolved.
Enable/settle 2.4V reference; verify gain/rate; keep PACE/RLD/WCT and RLD rerouting off.
F07/F08: protected body bias and wearer protection remain unimplemented.

V1.2 F12/F14 — PROCUREMENT REQUIREMENTS (not qualified installed parts)
R1–R32: 2k unchanged; 0.1%, >=0.1W, derating/pulse review required.
All 32 input 620pF caps: C0G/NP0, 1%; electrode impedance/CMRR still unverified.
C25/C51 VCAP3: 1uF unchanged, >=10V; qualify capacitance at ~5.2V DC.
C55 BP: 10nF unchanged; leakage <100nA at operating bias/temperature.
No MPN, footprint, physical performance or wearer-safety qualification is implied.

Version 1.2
McGill AI Lab — Researcher: Karen Chen Lai
Sheet: /
File: open-semg-16-v1.2.kicad_sch
Title: sEMG Wirstband
Size: A2 | Date: | Rev:
KiCad E.D.A. 10.0.7 | Id: 1/1